

CLASSIFICATION OF LOCALLY SYMMETRIC RIEMANNIAN SPACES

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ABSTRACT. We introduce locally and globally symmetric Riemannian spaces and explain É. Cartan's group theoretic approach to classifying all locally symmetric Riemannian spaces. In particular, we start with properties of geodesic symmetries and Cartan's criterion for local isometry and then move on to describe the structure of the identity component of the isometry group of a globally symmetric Riemannian manifold and demonstrate group representations of familiar symmetric manifolds. In the end, we state the classification theorem of irreducible orthogonal involutive Lie algebras (OIL-algebra) and explain how they answer the classification problem of locally symmetric Riemannian spaces.

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1. INTRODUCTION

A locally symmetric Riemannian space is a Riemannian manifold whose curvature tensor is invariant under all parallel translations, $\nabla R \equiv 0$. This definition is equivalent to having a strong local reflection symmetry. That is, around every point, geodesics may be reversed by a local isometry of the manifold. Classical examples of locally symmetric Riemannian spaces include Euclidean space, spheres, hyperbolic space and Lie groups with bi-invariant metrics. Symmetric spaces are commonly studied in representation theory, differential geometry and harmonic analysis.

É. Cartan initiated the study of symmetric spaces in 1926 and during 1930s he provided the full classification of locally symmetric Riemannian spaces. During his studies, he produced two different approaches to handle the classification problem. The first one is based on the so-called holonomy group which, for a Riemannian manifold M , is defined to be the group of all linear transformations of the tangent space $T_p M$ obtained by parallel translation along closed curves starting at $p \in M$. With this notion, he was able to reduce the classification problem into two simpler problems: 1) Classify all possible holonomy groups of symmetric spaces. 2) Determine the curvature tensor of a symmetric space by means of the holonomy group. He was able to answer the second problem but he did not complete the extensive calculations necessary for the solution of the first problem.

His second method relied heavily on Lie theory and enabled him to reduce the classification problem into the classification of simple Lie algebras over $\mathbb{R}$, which he had already solved in 1914. This method is based on the fact that for a locally symmetric Riemannian space M , geodesic symmetry with respect to each point is a local isometry. A beautiful consequence of this notion allows one to represent a globally symmetric Riemannian space M as a quotient G/K where G is a connected Lie group with an involutive automorphism σ whose fixed point set is (essentially) K .

This paper aims to provide basic theory of locally symmetric Riemannian spaces, and outline the classification of irreducible Orthogonal Involutive Lie algebras, explaining how they simplify the classification of locally symmetric Riemannian spaces.

Section 2 focuses on introducing the first definitions, and provides examples of locally and globally symmetric Riemannian spaces. Moreover, it explores Cartan's criterion for a local isometry.

Section 3 introduces transvection along curves, infinitesimal transvection at a point, Killing vector fields and explains the relationship of the Lie bracket of infinitesimal transvections at a point to the Riemannian curvature. We state the representation of a globally symmetric Riemannian space as a quotient G/K and explore the decomposition $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$ under a specific involutive automorphism of $\mathfrak{g}$ to associate each symmetric space an orthogonal involutive Lie algebra $(\mathfrak{g}, s, B)$. We demonstrate the quotient representations of and the orthogonal involutive Lie algebras corresponding to familiar symmetric spaces. Finally, we explain a method to obtain a simply-connected symmetric space M from a given abstract OIL-algebra.

Section 4 explores the classification of irreducible OIL-algebras. In particular, we state four different classes of irreducible OIL-algebras and explain the Cartan duality.

The locally symmetric Riemannian spaces are exactly the Riemannian manifolds whose universal cover is globally symmetric. Symmetric universal covers are simply-connected symmetric Riemannian manifolds. Hence, they are produced by OIL-algebras. The classification of irreducible OIL-algebras classifies all possible covering spaces of locally symmetric Riemannian spaces. Cartan's work continues to explore each class of OIL-algebra and the associated symmetric spaces however we end our discussion at the classification of irreducible OIL-algebras.

This paper assumes fundamental Riemannian geometry and Lie theory knowledge. The necessary Lie theoretical information is summarized in the 5th section.

2. LOCALLY SYMMETRIC RIEMANNIAN SPACES

Definition 2.1. Let (M, g) be a Riemannian manifold and $p \in M$. Consider a symmetric neighbourhood U_0 of $0 \in T_p M$ with respect to 0 that is diffeomorphic to its image under the exponential map. For example, you can take $U_0 = B(0, \epsilon) \subset T_p M$. A *geodesic symmetry* is a map $s_p : U_0 \rightarrow U_0$ taking $\exp(v)$ to $\exp(-v)$.

Proposition 2.2. Fix $p \in M$. The following are equivalent.

(i) The geodesic symmetry s_p at p is a local symmetry.

(ii) There exists a local isometry s of M defined on a neighborhood of p such that $s(p) = p$ and $ds_p = -id$.

(iii) There exists an involutive local isometry of M defined on a neighbourhood of p which has p as an isolated fixed point.

Proof. $i \implies ii$ Let $s = s_x$. We check $s(x) = s_x(x) = s_x(\exp_x(0)) = \exp_x(0) = x$. Now, let $v \in T_x M$ and consider the curve $\gamma(t) = \exp(tv)$. We have $\gamma(0) = x$ and $\gamma'(0) = v$, then we compute

$$\begin{aligned} ds_x(v) &= d(s_x)_x(v) \\ &= \left. \frac{d}{dt} \right|_{t=0} s_x(\exp(tv)) \\ &= \left. \frac{d}{dt} \right|_{t=0} \exp(-tv) \\ &= -v \end{aligned}$$

So we also have $ds_x = -id$.

$ii \implies i$ The map s in statement ii is already a local isometry so we only need to check whether it is a geodesic symmetry. Since s is an isometry, it satisfies the following

$$s \circ \exp_x(v) = \exp_{s(x)} \circ ds_x(v)$$

Using this, we see that

$$s(\exp_x(v)) = \exp_{s(x)}(ds_x(v)) = \exp_x(-v)$$

and so s is a geodesic symmetry.

$ii \implies iii$ Let s be the map given in statement ii and define $S = \{p \in M : s(p) = p\}$. Then S is a totally geodesic submanifold of M . $v \in T_x S \iff ds_x(v) = v$. Since $ds_x = -id$, we get that $T_x S = \{0\}$ and so x is a discrete point of S , ie an isolated fixed point of s . Since $s^2(p) = p$ and $(ds_p)^2 = d(s^2)_p = id$, we see that s^2 is identity on some neighbourhood of x .

iii $\implies$ *ii*. Assume s is a map given in statement *iii*. Then, $(ds_x)^2 = id$, therefore ds_x is an orthogonal transformation and the only possible eigenvalues of ds_x are $+1$ and -1 . If $+1$ is an eigenvalue of ds_x , then for an eigenvector v of $+1$, we would have $s(\exp(tv)) = \exp_x(ds_x(tv)) = \exp_x(tv)$ which contradicts with the fact that x is an isolated fixed point. Therefore, all eigenvalues must be -1 , hence proving $ds_x = -id$. $\square$

Definition 2.3. A Riemannian manifold (M, g) is called a *locally symmetric space* if the above proposition holds at every point of M . It is called a *globally symmetric space* if the geodesic symmetry s_p is globally defined on M and is an isometry for every $p \in M$.

Example 2.4. Euclidean space $\mathbb{R}^n$ is a globally symmetric space with the geodesic symmetry $s_x(x + v) = x - v$.

Example 2.5. Homogeneous spaces

M is called *homogeneous* if there exists a transitive group of isometries of M . We say that M is *locally homogeneous* if for all $x, y \in M$, there exists open neighbourhoods $U \ni x, V \ni y$ of M and an isometry $f : U \rightarrow V$ mapping x to y . Suppose M is locally homogeneous. Given an isometry $\gamma : [0, 1] \rightarrow M$ with $\gamma(0) = x, \gamma(1) = y$, we have

$$\gamma \circ s_x(\exp_x(v)) = s_y \circ \exp_y(d\gamma_x(v))$$

Hence, if s_x is a local isometry, then so is s_y . All of this shows that in a locally homogeneous Riemannian manifold M , it suffices to check that s_x is an isometry for at least one point $x \in M$ to conclude that M is a locally symmetric space.

Example 2.6. The unit sphere $\mathbb{S}^n$.

The orthogonal group $O(n+1)$ acts transitively on $\mathbb{S}^n$ by isometries. Moreover, given $x \in \mathbb{S}^n$, the restriction of the reflection with respect to the line $\mathbb{R}x$ onto $\mathbb{S}^n$ is a geodesic symmetry at x . By the previous example, we see that $\mathbb{S}^n$ is a symmetric space.

Example 2.7. Bi-invariant metric on Lie groups.

A Riemannian metric g on a Lie group G is *left-invariant* (or *right-invariant*) if the left (resp. right) translation map $L_g : G \rightarrow G$ (resp $R_g : G \rightarrow G$) given by $L_g(h) = gh$ (or $R_g(h) = hg$) is an isometry for every $g \in G$. A metric g is bi-invariant if it is both left-invariant and right-invariant. Recall that every $g \in G$ gives an automorphism of G via the conjugation map $\text{Inn}_g(h) = ghg^{-1}$. Then the differential of Inn_g at the identity induces the adjoint automorphism $d(G_g)_e : \mathfrak{g} \rightarrow \mathfrak{g}$. Let us also recall that the exponential map of G is the map $\exp^G : \mathfrak{g} \rightarrow G$ given by $\exp^G(v) = \gamma_v(t)$ where γ_v is the integral curve of v starting at e . This map coincides with the exponential map of G at identity. $\exp_e = \exp^G$.

Given a positive-definite inner product $\langle *, * \rangle$ on $\mathfrak{g} = T_e G$, we can take any basis $\{X_1, \dots, X_n\}$ of $\mathfrak{g}$ and extend it to a left-invariant frame on G by pushing forward these vectors $X_{i,g} = d(L_g)_e X_i$. In order to obtain a bi-invariant metric on G , we differentiate $R_g = L_g \circ \text{Inn}_{g^{-1}}$ at e to get

$$d(R_g)_e = d(L_g)_e \circ d\text{Ad}_{g^{-1}}$$

This shows that a left-invariant metric g is also right-invariant if the inner product $\langle *, * \rangle$ is Ad -invariant. Hence, an inner product on $\mathfrak{g}$ induces a bi-invariant metric on G if it is invariant under the adjoint action. One can also show that invariance under the adjoint is a sufficient condition for an inner product on $\mathfrak{g}$ to induce a bi-invariant metric.

We see that if G is a lie group endowed with a bi-invariant metric g , the map $L_{yx^{-1}}$ is an isometry of G sending x to y for any $x, y \in G$. Hence, G is homogeneous. We claim that the inverse map $\iota(x) = x^{-1}$ is a geodesic symmetry at $e \in G$. To check that ι is a geodesic symmetry, we will confirm that the second statement in Proposition 2.2 is satisfied by ι . Notice that $\iota(e) = e$.

Given $v \in \mathfrak{g} = T_e G$, the map $\gamma(t) = \exp^G(tv)$ is a curve in G with $\gamma(0) = e$ and $\gamma'(0) = v$. Therefore, we can compute

$$d\iota_e(v) = \left. \frac{d}{dt} \right|_{t=0} \iota(\exp^G(tv)) = \left. \frac{d}{dt} \right|_{t=0} \exp^G(-tv) = -v$$

Hence, $d\iota_e = -id$. Moreover, we have $\iota = R_{g^{-1}} \circ \iota \circ L_{g^{-1}}$, so

$$d\iota_g = d(R_{g^{-1}})_e \circ d\iota_e \circ d(L_{g^{-1}})_g$$

Since all of these maps are linear isometries, we see that $d\iota_g$ is a linear isometry for every $g \in G$. Hence, ι is a linear isometry as desired.

All of this show that G is a locally symmetric space.

We know that existence of involutive local isometries tell us that our manifold is locally symmetric. On the way to classifying locally symmetric Riemannian manifolds, one natural question is to ask when a linear isometry between tangent spaces gives rise to an isometry of manifolds. To make this notion precise, let us consider two manifolds $M, \widetilde{M}$ and two points $x \in M, \tilde{x} \in \widetilde{M}$. Suppose we are given a linear isometry $I : T_x M \rightarrow T_{\tilde{x}} \widetilde{M}$ and $V \subset M$ is a normal neighbourhood of x such that $\exp_x$ is defined on $I(\exp_x^{-1}(V))$. Let us lift I to a map between the two manifolds. That is, let's define $\varphi : V \rightarrow \widetilde{M}$ by

$$\varphi(y) = \exp_{\tilde{x}} \circ I \circ \exp_x^{-1}(y)$$

Finally, let's define the transformation induced by I between different tangent spaces of M and $\widetilde{M}$ as follows. For $y \in V$, let $P_\gamma : T_x M \rightarrow T_y M$ be the parallel translation along the geodesic $\gamma : [0, 1] \rightarrow M$ with $\gamma(0) = x, \gamma(1) = y$. Similarly, let $P_{\tilde{\gamma}} : T_{\tilde{x}} \widetilde{M} \rightarrow T_{\tilde{y}} \widetilde{M}$ be the parallel translation along the geodesic $\tilde{\gamma} := \varphi \circ \gamma$. Then define $I_\gamma : T_y M \rightarrow T_{\tilde{y}} \widetilde{M}$ by

$$I_\gamma(v) = P_{\tilde{\gamma}} \circ I \circ P_\gamma^{-1}(v)$$

Cartan was interested in understanding when a linear isometry I of tangent spaces could be extended into a local isometry φ between manifolds and his answer was the following theorem.

Theorem 2.8. (*Cartan's Theorem*) Let φ, I, I_γ be defined as above. If, for every $y \in V$ and for every $u, v, w \in T_y M$ we have

$$I_\gamma(R(u, v)w) = \tilde{R}(I_\gamma(u), I_\gamma(v))I_\gamma(w)$$

where R and $\tilde{R}$ are the curvature tensors of M and $\widetilde{M}$, then φ is a local isometry and $d\varphi_y = I_\gamma$ for every $y \in V$.

Proof sketch. We have the following figure.

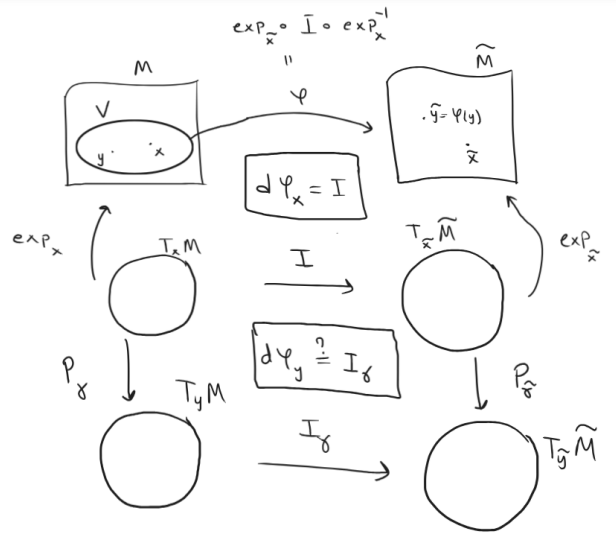


FIGURE 1

It suffices to prove that given $y \in V, v \in T_y M$, one has $\|d\varphi_y(v)\| = \|v\|$. Consider the Jacobi field along the geodesic $\gamma : [0, 1] \rightarrow M$ connecting $x = \gamma(0)$ to $y = \gamma(1)$ with initial conditions $J(0) = 0, J(1) = v$. We know that it is given by $J(t) = d(\exp_x)_{t\gamma'(0)}(tw)$ where $w = J'(0)$. (This is Lemma 10.7 in [Lee97].) From this we obtain $v = J(1) = d(\exp_x)_{\gamma'(0)}(w)$ and

$$d\varphi_y(v) = d(\exp_{\tilde{x}})_{\tilde{\gamma}'(0)}(I(w)) = \tilde{J}(1)$$

where $\tilde{J}(t)$ is the Jacobi field along the geodesic $\tilde{\gamma} = \varphi \circ \gamma$ with initial conditions $\tilde{J}(0) = 0$ and $\tilde{J}'(0) = I(w) = I(J'(0))$. We can pick an orthonormal frame $E_1, \dots, E_n$ of parallel vector fields along γ and write

$$J(t) = \sum_i \alpha_i E_i(t)$$

The coefficients satisfy a particular second order ODE. Finally, we can do the same for $\tilde{\gamma}$ and write

$$\tilde{J}(t) = \sum_j \beta_j \widetilde{E}_j(t)$$

where $\widetilde{E}_1, \dots, \widetilde{E}_n$ is an orthonormal frame of parallel vectors along $\tilde{\gamma}$. Since $\widetilde{E}_j(t) = I_{\gamma(t)}(E_j(t))$ and $I_{\gamma(t)}(\gamma'(t)) = \tilde{\gamma}'(t)$, we see that $\alpha_i = \beta_i$ by the hypothesis of the theorem and as a result we get $\|d\varphi_y(v)\| = \sum_i \beta_i^2(1) = \sum_i \alpha_i^2(1) = \|v\|$. $\square$

Equipped with the power of this theorem he was able to give a useful characterization of locally symmetric spaces in the form of the following theorem.

Theorem 2.9. *A Riemannian manifold M is a locally symmetric space if and only if $\nabla R \equiv 0$.*

Let us also note that the above theorem is equivalent to requiring the sectional curvature of M to be invariant under all parallel transports of tangent planes.

3. GLOBALLY SYMMETRIC SPACES

In this section, we start exploring the relationship between Lie groups and connected Riemannian manifolds. The first question we are interested in is the structure of the set of isometries of a connected Riemannian manifold M . Let (M, g) denote a connected Riemannian manifold for the rest of this section.

Myers-Steenrod theorem states that the isometry group G of a connected Riemannian manifold M has Lie group structure when endowed with the topology called the *compact open topology*. Moreover, the action of G on M is smooth and represents its Lie algebra $\mathfrak{g}$ as the Lie algebra of Killing vector fields on M , ie vector fields on M whose flow is a local isometry of M .

Definition 3.1. Fix $x \in M$ and consider a curve $\gamma : (-\varepsilon, \varepsilon) \rightarrow M$ with $\gamma(0) = x$. Let s_t denote the geodesic symmetry $s_{\gamma(t)}$. A *transvection along γ* is a local isometry given by $p_t = s_{t/2}s_0$. It is easy to check that p_t induces the translation $p_t(\gamma(t_0)) = \gamma(t_0 + t)$ and if $v \in T_{\gamma(t_0)}M$, then

$$X(t) = d(p_{t-t_0})_{\gamma(t_0)}(v)$$

is a parallel vector field along γ . Moreover, $\{p_t\}$ is a local 1-parameter group of local isometries of M , ie $p_{t+t'} = p_t p_{t'}$ whenever both sides are defined.

Definition 3.2. Fix $x \in M$. Then, each 1 parameter group of isometries $\{p_t\}$ gives rise to a vector field Y along some geodesic γ defined by setting $Y(y) = \frac{d}{dt}|_{t=0} p_t y$. Defined this way, $Y(y)$ simply becomes the tangent vector at y to the geodesic connecting x to y . Such a vector field Y is called an *infinitesimal transvection at x* .

Definition 3.3. Let G be a the group of isometries of a Riemannian manifold M . The *isotropy group at x* is the Lie subgroup of G consisting of elements that act trivially on x , ie the set $\{g \in G : g \cdot x = x\}$

We note the following important relation between the curvature tensor and the bracket of the set of infinitesimal transvections at x .

Theorem 3.4. *Let M be a symmetric space and let $p \in M$. Let $\mathfrak{p}$ the set of infinitesimal transvections at p . Then, for all $U, V, W \in \mathfrak{p}$ we have*

$$R(U, V)W = [U, [V, W]]$$

A proof of this theorem can be found in Chapter 4 of [Hel77].

We can now introduce the most important theorem of this paper.

Theorem 3.5. *Let M be a globally symmetric space with Lie group of isometries $\text{Iso}(M)$. Let G be the identity component of $\text{Iso}(M)$. Fix $x \in M$ and let K denote the subgroup of G which leaves x fixed and let P be the set of transvections at x . Let $\mathfrak{g}, \mathfrak{k}$ denote the Lie algebras of G and K , respectively. Finally, let $\mathfrak{p}$ denote the set of infinitesimal transvections at x . Then,*

(i) $M = G/K$.

(ii) $s = \text{Ad}_{s_x}$ defines an involutive isomorphism of $\mathfrak{g}$ and $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$ is the ± 1 eigenspace decomposition.

(iii) The projection $\pi : G \rightarrow M$ is given by $\pi(g) = gx$ has a differential $d\pi_e : T_e G = \mathfrak{g} \rightarrow T_x M$ which satisfies $\ker d\pi_e = \mathfrak{k}$. As a result, $\mathfrak{g}/\mathfrak{k} = \mathfrak{p} \cong T_x M$.

(iv) If $k \in K$ and $Y \in \mathfrak{p}$, then $\pi_*(\text{Ad}_k Y) = dk_x(\pi_* Y)$. In particular, the inner product on $T_x M$ lifts to an Ad_K -invariant inner product on $\mathfrak{p}$.

Proof of the this theorem can be found in [Gor21], Proposition 1.5.1.

As a result, we associate a triple $(\mathfrak{g}, s, B)$ to each globally symmetric connected Riemannian manifold M where $\mathfrak{g}$ is the Lie algebra of the identity component G of the isomorphism group of M , $s = \text{Ad}_{s_x}$ is an involutive isomorphism of $\mathfrak{g}$ - where $x \in M$ is any base-point - and B is the inner product on $\mathfrak{g}$ which lifts to an inner product on the vector subspace $\mathfrak{p}$ that is invariant under the adjoint action of the isotropy group of x . Such a triple is called an *orthogonal involutive Lie algebra of M at x* .

Example 3.6. The Unit Sphere S^n The group of isometries of S^n is the orthogonal group $O(n+1)$, its identity component is $SO(n+1)$ and the lie algebra of $SO(n+1)$ is $\mathfrak{g} = \mathfrak{so}(n+1)$. Let us assume $x = (1, \dots, 0) \in S^n$ is the north pole. Then the isotropy group at x is $O(n)$. The stabilizer inside $SO(n+1)$ is $SO(n)$. Thus we have $S^n \cong SO(n+1)/SO(n)$. The geodesic symmetry s_x sends a point $(x^1, \dots, x^{n+1}) \in S^n$ to the point $(-x^1, x^2, \dots, x^{n+1})$. Therefore we see that $s = \text{Ad}_{s_x} : \mathfrak{so}(n+1) \rightarrow \mathfrak{so}(n+1)$ is given by

$$s(A) = \begin{pmatrix} -1 & 0 \\ 0 & I_n \end{pmatrix} A$$

whose $+1$ and -1 eigenspaces decompose as

$$\mathfrak{k} = \left\{ \begin{pmatrix} 0 & 0 \\ 0 & A \end{pmatrix} : A \in \mathfrak{so}(n) \right\} \quad \mathfrak{p} = \left\{ \begin{pmatrix} 0 & -v^t \\ v & 0 \end{pmatrix} : v \in \mathbb{R}^n \right\}$$

The inner product on $\mathfrak{so}(n+1)$ is given by $B'(X, Y) = C \cdot \text{tr}(X^t Y)$ for some constant C . This inner product comes from a negative Killing form. So the orthogonal involutive Lie algebra associated to S^n is $(\mathfrak{so}(n+1), s, B)$.

Example 3.7. Real Projective Space $\mathbb{R}P^n$ We know that $SO(n+1)$ acts transitively on the unit sphere, and $O(n)$ is isomorphic to the stabilizer of a pair of antipodal points $\{\pm x\}$. Since $\mathbb{R}P^n \cong S^n/(x \sim -x)$ the isotopy group at $[x] \in \mathbb{R}P^n$ is exactly $O(n)$. Hence, we see that $\mathbb{R}P^n \cong SO(n+1)/O(n)$.

Example 3.8. Complex Projective Space $\mathbb{C}P^n$ $U(n+1)$ acts transitively on $\mathbb{C}P^n$. Since $U(n+1)$ is connected, its identity component is itself. A unitary matrix g stabilizes the line $l = \mathbb{C}e_1$ if and only if $ge_1 = ze_1$ for some $z \in \mathbb{C}$, in which case it must be of the form

$$g = \begin{pmatrix} z & 0 & \dots & 0 \\ 0 & & & \\ \vdots & & A & \\ 0 & & & \end{pmatrix}$$

where $A \in U(n)$. Then the isotropy group of an element of $\mathbb{C}P^n$ is isomorphic to $U(1) \times U(n)$. Therefore, we have

$$\mathbb{C}P^n \cong U(n+1)/U(1) \times U(n)$$

The lie algebra of $U(n+1)$ is the set of $(n+1) \times (n+1)$ skew-Hermitian matrices $\mathfrak{u}(n+1)$. Let $x = \mathbb{C}e_1$. The geodesic symmetry s_x is a unitary matrix that fixes x and acts as the negative identity on the complement

of the line x in $\mathbb{C}^n$. This shows that the involutive isomorphism $s : \mathfrak{u}(n+1) \rightarrow \mathfrak{u}(n+1)$ is given by

$$s(A) = \begin{pmatrix} -1 & 0 \\ 0 & I_n \end{pmatrix} A$$

Then we have

$$\mathfrak{k} = \mathfrak{u}(1) \oplus \mathfrak{u}(n) \quad \mathfrak{p} = \left\{ \begin{pmatrix} 0 & -v^* \\ v & 0 \end{pmatrix} : v \in \mathbb{C}^n \right\}$$

At this point we can use Theorem 3.5 to obtain more information about $\mathbb{C}P^n$. Let $X, Y, Z \in \mathfrak{p}$ with corresponding $x, y, z \in \mathbb{C}^n$. We obtain a vector $r(x, y)z \in \mathbb{C}^n$ corresponding to $R(X, Y)Z = [X, [Y, Z]] \in \mathfrak{p}$. Computing the bracket gives

$$r(x, y)z = z(y^*x - x^*y) + x(y^*z) - y(x^*z) \in \mathbb{C}$$

For $y = z$, we get

$$r(x, y)y = y(y^*x - 2x^*y) + x(y^*y)$$

where $y^*x = \langle x, y \rangle_c$ is the usual Hermitian inner product of $x, y \in \mathbb{C}$. Let y be a unit vector and x perpendicular to y with respect to the real inner product $\langle x, y \rangle = \Re(\langle x, y \rangle_c)$. Since we have $y^*x = -x^*y = i\langle x, iy \rangle$ we see that

$$r(x, y)y = x + 3\langle x, iy \rangle iy$$

This shows that the eigenvalues of $r(\cdot, y)y$ on $y^\perp$ are 1 on $\{y, iy\}^\perp$ and 4 on $\mathbb{R}iy$. Therefore, the sectional curvature of $\mathbb{C}P^n$ varies between 1 and 4. Furthermore, the value 1 is taken on any real plane spanned by x, y with $x \perp y, iy$ and the value 4 is taken by any complex plane spanned by y, iy .

We end this section by describing a procedure to construct a simply-connected symmetric Riemannian manifold M from a given orthogonal involutive Lie algebra $(\mathfrak{g}, s, B)$. Let G be the simply-connected Lie group whose Lie algebra is $\mathfrak{g}$. Let $K \subset G$ be the connected subgroup with Lie algebra $\mathfrak{k}$. There is an involution $\sigma : G \rightarrow G$ such that $d\sigma = s$ and K is the identity component of the fixed point subgroup $G^\sigma = \{g \in G : \sigma(g) = g\}$, hence K is closed in G . Then $M = G/K$ is a simply connected homogeneous space. However, the action of G on M might not be effective. That is, there could be a non-trivial element $g \in G$ that acts as the identity on M . This happens when $g \cdot hK = hK, \forall h \in G \implies g \in \bigcap_{h \in G} hKh^{-1}$. Therefore, let $Z = \{g \in G : g \cdot M = id\}$ be the kernel of the action. Then, setting $M = (G/Z)/(K/Z)$ ensures that G acts effectively on M . Moreover, $G' := G/Z, K' := K/Z$ have Lie algebras $\mathfrak{g}, \mathfrak{k}$. The projection $\pi : G' \rightarrow M$ is a map whose image is isomorphic to $\mathfrak{p} \cong T_x M$ where $x = 1K'$ and B defines a K' -invariant inner product on $T_x M$ which extends to a G' -invariant Riemannian metric on M . Moreover, σ induces a map $\psi : M \rightarrow M$ by the rule $\psi(gK') = \sigma(g)K'$. We can also express ψ as $\psi(gx) = \sigma(g)x = \pi\sigma(g)$. Then, ψ fixes x and satisfies $d\psi_x(Y_x) = \pi_*s(Y) = -Y_x$ for $Y \in \mathfrak{p}$. So $d\psi_x = -id$. Since $\psi(gg_1K) = \sigma(gg_1)K = \sigma(g)\sigma(g_1)K = \sigma(g)\psi(g_1K)$, we have $\psi \circ g = \sigma(g) \circ \psi$. Then differentiating gives

$$d\psi_{gx} = d(\sigma(g))_x \circ d\psi_x \circ d(g^{-1})_{gx}$$

for every $g \in G'$. This shows that ψ is an isometry of M with $d\psi_x = -id$. Therefore, M is a symmetric space.

4. CLASSIFICATION OF IRREDUCIBLE ORTHOGONAL INVOLUTIVE LIE ALGEBRAS

Throughout this section, let M be a globally symmetric Riemannian manifold with associated orthogonal involutive algebra (OIL-algebra) $(\mathfrak{g}, s, B)$. Let G denote the identity component of the isometry group of M and let's write $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$ under s .

Proposition 4.1. *If G is semisimple, then it is generated by the transvections of M at the basepoint.*

It follows from this lemma and Theorem 3.4 that a symmetric space M with a semisimple isometry group G has the representation $M = G/K$ where K is the isotropy group at a point.

Definition 4.2. An OIL-algebra $(\mathfrak{g}, s, B)$ is called *Euclidean* if $[\mathfrak{p}, \mathfrak{p}] = 0$ and it is called *irreducible* if it is not Euclidean and the adjoint representation of $\mathfrak{k}$ on $\mathfrak{p}$ is irreducible.

We have the following decomposition theorem.

Proposition 4.3. *Let $(\mathfrak{g}, s, B)$ be an OIL-algebra. Then there is a decomposition*

$$\mathfrak{g} = \mathfrak{g}_0 \oplus \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_r$$

into a direct sum of ideals such that:

(i) Each $\mathfrak{g}_i$ is s -invariant.

(ii) $(\mathfrak{g}_0, s|_{\mathfrak{g}_0}, B|_{\mathfrak{p}_0})$ is Euclidean.

(iii) $(\mathfrak{g}_i, s|_{\mathfrak{g}_i}, B|_{\mathfrak{p}_i})$ is irreducible for $i = 1, \dots, r$.

It can also be shown that the above decomposition is unique up to permutation of the simple factors. It follows as a corollary of the above proposition that if $\mathfrak{g}$ is simple then $(\mathfrak{g}, s, B)$ is irreducible.

The name Euclidean is not random. In fact, the only Euclidean OIL-algebras are of the form $(\mathfrak{g}, s, B)$ where $\mathfrak{g} = \mathfrak{k} + \mathbb{R}^n$ (semidirect sum), $\mathfrak{k} \subset \mathfrak{so}(n)$, s is $+1$ on $\mathfrak{k}$ and -1 on $\mathbb{R}^n$ and B is the standard inner product on $\mathbb{R}^n$.

Theorem 4.4 (Classification). *Let β denote the Killing form of $\mathfrak{g}$. The irreducible OIL-algebras $(\mathfrak{g}, s, B)$ fall into four pairwise disjoint classes as follows:*

Class 1: $\mathfrak{g}$ is a compact simple Lie algebra, s is an involutive automorphism of $\mathfrak{g}$ and $B = \lambda\beta$ with $\lambda < 0$.

Class 2: $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{h}$ where $\mathfrak{h}$ is a compact simple Lie algebra, $s(X, Y) = (Y, X)$ for $(X, Y) \in \mathfrak{g}$ and $B = \lambda\beta$ with $\lambda < 0$.

Class 3: $\mathfrak{g}$ is a non-compact simple Lie algebra with $\mathfrak{g}^c$ simple, $\mathfrak{k}$ a maximal compact subalgebra and $B = \lambda\beta$ with $\lambda > 0$.

Class 4: $\mathfrak{g}$ is the realification of a compact simple Lie algebra, $\mathfrak{k}$ is a compact real form and a maximal compact subalgebra, s is complex conjugation over $\mathfrak{k}$, and $B = \lambda\beta$ with $\lambda > 0$.

A proof of this theorem can be found in [Gor21].

We know that B is a Ad_K -invariant inner product on $\mathfrak{p} \cong T_x M$. The restriction of the Killing form β onto $\mathfrak{p}$ is another K -invariant symmetric bilinear form on $\mathfrak{p}$. Since K acts irreducibly on $\mathfrak{p}$, these symmetric bilinear forms differ only by a factor λ . That is, $B = \lambda\beta$. The sign of λ determines the type of the symmetric space. If $\lambda = 0$, then M has zero curvature, ie is a flat space. If $\lambda < 0$, then M is called of *compact type* and if $\lambda > 0$, then M is called of *non-compact type*. Classes 1 and 2 correspond to symmetric spaces of compact type and classes 3 and 4 correspond to symmetric spaces of non-compact type.

As a final note, we explain the Cartan duality. From any Lie algebra $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$, we obtain another Lie algebra $\mathfrak{g}'$ on the same set as $\mathfrak{g}$ by defining the Lie bracket as

$$[a, b]' = [a, b], [a, x]' = [a, x], [x, y]' = -[x, y]$$

for every $x, y \in \mathfrak{p}$ and for every $a, b \in \mathfrak{k}$. The new orthogonal involutive Lie algebra obtained from $\mathfrak{g}'$ corresponds to a symmetric space M whose curvature tensor has the opposite sign for that of the symmetric space corresponding to the original OIL-algebra. This construction allows one to obtain a non-compact type symmetric space from a compact type and vice versa. In particular, Cartan duality exchanges classes 1 and 3, 2 and 4.

As mentioned in the introduction, the classification of OIL-algebras achieves the classification of all possible simply-connected symmetric spaces M which are the universal covering maps of locally symmetric Riemannian spaces.

At this point, the classification of locally symmetric Riemannian spaces have fully entered to the realm of Lie theory, which we do not explore any further in this paper. The classification is continued by exploring compact simple Lie algebras and their involutive automorphisms for classes 1 and 2. Cartan duality is then used to explore classes 3 and 4.

5. FACTS FROM LIE THEORY

In this section, we provide a review of the necessary (and minimal) Lie theory knowledge that will help the reader follow the paper easily.

A *Lie group* is a group G that has a manifold structure such that addition and inverse are smooth maps. A *Lie algebra* is a vector space $\mathfrak{g}$ with a bilinear product $[\cdot, \cdot] : \mathfrak{g} \rightarrow \mathfrak{g}$ satisfying

$$[a, [b, c]] + [b, [c, a]] + [c, [a, b]] = 0 \quad (\text{Jacobi identity})$$

This map is called the *Lie bracket*.

It turns out that the tangent space at the identity of a Lie group G has a Lie algebra structure. The Lie algebra of the Lie group G is then defined to be $\mathfrak{g} = T_e G$. Left and right multiplication maps (which will be denoted by L_g, R_g) on G are diffeomorphisms on G whose differentials provide an isomorphism between $T_a G$ and $\mathfrak{g}$ for any $a \in G$.

A vector field X on G is called left-invariant if $d(L_g)_h X_h = X_{gh}$. There is a bijection between the set of left-invariant vector fields on G and $\mathfrak{g}$ given obtained pushing forward the vectors in $\mathfrak{g}$ by $d(L_g)_e : \mathfrak{g} \rightarrow T_g G$. Right-invariant vector fields are defined similarly.

A *1-parameter subgroup of a Lie group* G is a Lie group morphism

$$\gamma : \mathbb{R} \rightarrow G$$

It can be proven that for a Lie group G with Lie algebra $\mathfrak{g}$, the left-invariant vector fields are complete, ie integral curves exist for all time. Moreover, the unique integral curve $\gamma(t)$ of a left invariant vector field X with initial condition $\gamma(0) = e$ is a 1-parameter subgroup of G and the flow of this vector field is given by $\Phi_t(g) = g\gamma(-t)$.

The *exponential map* for a Lie group G is the smooth map defined by

$$\exp^G : \mathfrak{g} \rightarrow G, \exp^G(\xi) = \gamma^\xi(1)$$

where $\gamma^\xi(t)$ is a 1-parameter subgroup with $\dot{\gamma}^\xi(0) = \xi$. G can be viewed as a Riemannian manifold when it is equipped with a Riemannian metric. In that case, $\exp^G$ becomes the same as the exponential map at the identity $\exp_e$. In particular, $d\exp_e^G = id$.

The *adjoint representation of a Lie algebra* $\mathfrak{g}$ is a Lie algebra homomorphism (a map that respects the Lie bracket) defined by

$$\text{ad} : \mathfrak{g} \rightarrow \text{gl}(\mathfrak{g}), (\text{ad } x)(y) = [x, y]$$

The *Adjoint action* of a Lie group G on itself is a Lie group homomorphism defined by

$$\text{Ad} : G \rightarrow \text{Aut}(G), a \mapsto \text{Ad}_a$$

where $\text{Ad}_a : G \rightarrow G$ is conjugation by a , ie $\text{Ad}_a(b) = aba^{-1}$.

Given an automorphism C of G , one can take the differential at the identity to obtain an automorphism of $\mathfrak{g}$. Applying this to the adjoint action of G on itself gives us an automorphism $d\text{Ad}_e \in \text{Aut}(\mathfrak{g})$.

The *adjoint representation of the Lie group* G on $\mathfrak{g}$ is the map defined by

$$\text{Ad} : G \rightarrow \text{Aut}(\mathfrak{g}), g \mapsto \text{Ad}_g = d(\text{Ad}_g)_e$$

There is a strong connection between these three different maps. Namely, the adjoint representation $\text{ad} : \mathfrak{g} \rightarrow \text{Der}(\mathfrak{g})$ is the differential of the adjoint representation $\text{Ad} : G \rightarrow \text{Aut}(\mathfrak{g})$. In particular, $\exp^G(\text{ad}_\xi) = \text{Ad}(\exp^G \xi)$.

The *Killing form* of a finite dimensional Lie algebra $\mathfrak{g}$ is a symmetric bilinear form

$$\kappa(\xi, \eta) = \text{tr}_{\mathfrak{g}}(\text{ad}_\xi \text{ad}_\eta)$$

The Killing form is invariant under all automorphisms of $\mathfrak{g}$. It is also ad -invariant and, if $\mathfrak{g}$ is the Lie algebra of some Lie group G , it is Ad -invariant.

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